

Product Requirements Document (PRD): AI Job Application Automation Platform

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1. Executive Summary

Product Name (Working Title)

ApplyAI

Vision

Build an AI-powered career assistant that enables users to upload a resume once, automatically discover relevant job opportunities, generate personalized application materials, and—with the user's authorization—submit applications or prepare outreach to recruiters across supported job platforms. The product should function as a complete AI job search assistant rather than simply an auto-apply bot.

2. Problem Statement

Job seekers spend significant time on repetitive tasks such as:

- Searching across multiple job portals
- Filling out identical application forms
- Uploading resumes repeatedly
- Writing similar cover letters
- Tracking application status
- Following up with recruiters

These repetitive tasks reduce the time available for networking, interview preparation, and skill development.

3. Goals

Primary Goals

- Reduce job application time by 90%
- Increase relevant job matches using AI
- Automate repetitive application workflows
- Improve ATS compatibility of resumes
- Provide a centralized application dashboard

Business Goals

- Acquire 100,000 users
- Achieve 10% paid conversion
- Build integrations with major ATS platforms
- Become an AI career assistant platform

4. Target Users

User Type	Needs & Pain Points
Students	Internship search, entry-level jobs, resume improvement.
Experienced Professionals	Higher salary roles, remote jobs, career growth.
Freelancers	Contract work, remote opportunities, global clients.
Career Switchers	Skill matching, resume rewriting, transferable skills identification.

5. User Journey

1. User Signup
2. Login
3. Upload Resume
4. AI Resume Parsing
5. Candidate Profile Created
6. Job Search
7. AI Matching
8. Resume Customization
9. Application Review
10. Apply (User-approved)
11. Dashboard Tracking

6. Core Features

Module 1: Authentication

Features: Email login, Google login, LinkedIn login (if supported), Password reset, Multi-factor authentication (optional).

Acceptance Criteria: Secure authentication, JWT/session management, Role-based access.

Module 2: Resume Upload

Supported Formats: PDF, DOCX (Max size: 10 MB).

AI Extracts: Name, Email, Phone, Skills, Experience, Education, Certifications, Projects, Languages.

Output: Structured Candidate Profile.

Module 3: AI Candidate Profile

Example Profile Payload:

```
{  
  "skills": ["Python", "React", "AWS"],  
  "experience": 5,  
  "preferred_location": "Remote",  
  "expected_salary": "18 LPA",  
  "work_authorization": "India"  
}
```

Features: Editable profile, AI suggestions, Missing skills detection, ATS score.

Module 4: Job Discovery

Sources: LinkedIn, Indeed, Wellfound, Greenhouse, Lever, Company career pages, RemoteOK, We Work Remotely (subject to terms and APIs).

Filters: Location, Remote, Salary, Experience, Company, Skills, Employment Type.

Module 5: AI Job Matching

Every job receives a compatibility score breakdown:

Parameter	Score
Skills	95%
Experience	90%
Education	80%
Location	100%
Salary	90%
Overall Match	93%

Module 6: Resume Optimization

AI Generates: ATS Resume, Cover Letter, Professional Summary based on Resume and Job Description inputs.

Module 7: Application Automation

For supported platforms and with explicit user authorization: Open Job → Complete Form → Upload Resume → Generate Answers → Submit → Save Confirmation.

Module 8: Recruiter Outreach

Suggests public recruiter contacts, generates outreach/follow-up emails, and tracks email

status.

Module 9: Dashboard

Statistics: Applications Sent, Pending, Interviews, Offers, Rejections.

Timeline: Applied → Viewed → Interview → Offer → Accepted.

Module 10: Notifications

Alerts for new matching jobs, interview requests, recruiter replies, profile tips, and weekly summaries.

Module 11: AI Career Coach

Provides resume reviews, interview preparation, salary guidance, skill gap analysis, and learning roadmaps.

7. AI Architecture Sequence

Resume Upload → Resume Parser → LLM Information Extraction → Candidate Knowledge Base → Job Search Engine → Embedding Similarity → LLM Ranking → Resume Generator → Application Engine → Analytics Dashboard.

8. Technical Architecture Stack

Layer	Technologies
Frontend	Next.js, React, Tailwind CSS, TypeScript
Backend	FastAPI (Python), REST API, GraphQL (optional)
Database	PostgreSQL (Users, Profiles, Resumes, Jobs, Applications, Emails, Companies)

Layer	Technologies
AI Stack	OpenAI API, Embeddings, Resume Parser, OCR (for scanned resumes)
Infrastructure	Docker, Kubernetes, Redis, AWS S3 / Cloudflare R2, GitHub Actions, Cloudflare CDN
Authentication & Email	Clerk, Auth0, Firebase Auth Gmail OAuth, Microsoft Outlook, SendGrid, Amazon SES

9. API Overview

- **User:** POST /signup, POST /login, GET /profile, PUT /profile
- **Resume:** POST /resume/upload, GET /resume, PUT /resume
- **Jobs:** GET /jobs, GET /jobs/recommended, POST /jobs/search
- **Application:** POST /apply, GET /applications, DELETE /application

10. Core Database Schema Fields

- **Users:** id, name, email, password, created_at
- **Profiles:** id, user_id, skills, experience, education, location
- **Applications:** id, user_id, job_id, status, applied_at
- **Jobs:** id, title, company, salary, location, url

11. Non-Functional Requirements

- **Performance:** Resume parsing < 5s, Job search < 3s, Dashboard load < 2s.
- **Availability:** 99.9% uptime.
- **Security:** HTTPS, Encrypted file storage, OAuth, JWT/session authentication, GDPR-ready privacy controls, Audit logs.

12. Success Metrics (KPIs)

User Metrics: DAU, MAU, Resume uploads, Applications submitted, Interview invitations, Offer rate.

Business Metrics: Subscription conversion, Churn rate, CAC, LTV.

13. Risks & Compliance

- **Technical Risks:** Platform web layout changes, API rate limits, automation failures, duplicate applications.
- **Legal & Compliance:** Respect platform terms of service, use official APIs, mandatory explicit user consent, strict encryption, user toggle to disable automated features.

14. Product Roadmap Phases

1. **Phase 1 (MVP - 8–10 weeks):** Auth, resume upload/parsing, AI profile, job discovery & matching, dashboard.
2. **Phase 2 (6–8 weeks):** Resume/cover letter customization, email outreach, application tracking, notifications.
3. **Phase 3 (10–12 weeks):** Automation workflows, AI interview coach, salary insights, analytics, browser extension, mobile app.

15. Future Enhancements

- AI voice interview prep & multi-language resume support
- Portfolio website generator & GitHub profile analysis
- LinkedIn profile optimization & referral matching
- AI networking & salary negotiation assistants
- Enterprise university and recruitment partnerships

16. Conclusion

ApplyAI is structured to simplify end-to-end job discovery and application processing cleanly. By maintaining a user-first, highly controlled framework, the system maximizes application accuracy and complies with platform regulations, paving the way for scalable AI career assistance.